AUTONOMOUS SOFTWARE EXPERIMENTS ON HERA: TECHNICAL AND OPERATIONAL REQUIREMENTS

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1. TECHNICAL REQUIREMENTS

To ensure the software can fly safely on Hera and to keep the safety guarantees of the Core 1 sandbox, all Experimental Software (ESW) must comply with the requirements below.

1.1. Target platform and toolchain

- **Processor:** ESW shall target the GR712RC LEON3 architecture.
- **Compiler:** ESW shall be compiled with Frontgrade Gaisler Bare C Cross Compiler System for LEON3 GCC, BCC 4.4.2 1.0.52.

Proposers should assume a constrained embedded environment and choose algorithms, data formats, and implementation approaches that are suitable for this processor and toolchain.

1.2. Execution model

- **No operating system:** ESW shall not use any operating system services. It shall run in a bare metal style within the provided experimental framework.

The experiment should be a self-contained application, with explicit control of its execution flow, timing, and internal state (using only the platform interfaces provided).

1.3. Language and standards compliance

- **Language:** ESW source code shall be written in C.
- **Standards:** ESW shall follow MISRA rules and comply with ECSS E ST 40C requirements applicable to Category D software.

Proposals should briefly explain how compliance will be achieved and shown (for example, by static analysis reports and documented deviations if needed).

1.4. Memory management constraints

- **No dynamic allocation:** `malloc/free` (and similar) are forbidden. ESW shall use static allocation, or a fixed pool allocator initialised at startup.

Proposals should include a clear memory approach: expected buffer sizes, any fixed pools, and how worst-case memory usage is bounded.

1.5. Runtime and library restrictions

- **No system calls:** ESW shall not use system calls.

- **No external libraries:** ESW shall not rely on external libraries. Only ESA's LibmCS math library may be used.

1.6. Hardware access and I/O

ESW shall not access spacecraft hardware directly. All input/output must use the inter-core Communication API provided by the platform. Interface details are in Annex A, and Data Pool parameters are described in Annex B.

2. INPUTS AVAILABLE TO THE ESW

2.1. Mission Data Pool

The ESW can read a selected set of spacecraft parameters from the spacecraft data pool. The data pool contains observable parameters that are updated by Hera's on-board software and include data coming from different units and subsystems. Access is read-only. A list of available parameters is provided in Annex B.

If your experiment requires spacecraft state data, list the parameter categories you would use and explain how your algorithm would use them.

2.2. Payload data sources

The ESW can request data acquisitions from specific payloads using the platform interfaces. The available options and the on-board processing possibilities depend on the instrument, as described below.

2.2.1. AFC — Asteroid Framing Camera

The Asteroid Framing Camera (AFC) is Hera's main visible-light camera and supports both navigation and science. It provides meter-scale images for asteroid mapping (shape, surface features, DART crater) and supports autonomous navigation. AFC is the main option for in-flight image-processing experiments (feature extraction, tracking, compression, prioritisation).

- **Detector:** 1020×1020 pixel FaintStar2 CMOS sensor.
- **Optics:** $5.5^\circ \times 5.5^\circ$ field of view.
- The ESW can request an AFC image. The image is available for on-board processing by the ESW and can also be requested for downlink in the next communications window.

2.2.2. TIRI — Thermal Infrared Imager

TIRI (provided by JAXA) supports thermal mapping of the Didymos–Dimorphos system to estimate properties such as thermal inertia, porosity, and surface roughness, supporting analysis of the DART impact aftermath.

- **Detector:** 1024 $\times$ 768 uncooled micro-bolometer.
- **Wavelength range:** 7–14 μm.
- **Filters:** 8-position wheel (one wide-band + six narrow-band filters).
- The ESW can request TIRI images; they will be downlinked in the next communication window. Making TIRI images available for on-board processing in the ESW is still being assessed.

2.2.3. HyperScout — Hyperspectral Imager

HyperScout is a visible-to-near-infrared hyperspectral instrument used to map mineralogy and monitor the dust environment.

- **Detector:** 2048 $\times$ 1088 CMOS sensor.
- **Optics/bands:** $\sim$15.5° $\times$ 8.3° field of view, 25 spectral bands (650–950 nm).
- The ESW can request HyperScout images. They will be downlinked in the next communication window. HyperScout images cannot be processed on-board by the ESW.

HyperScout is suitable for experiments that trigger/schedule acquisitions, or correlate spacecraft state with downlinked science. It is not suitable for experiments that require on-board hyperspectral processing.

2.2.4. PALT — Planetary Altimeter

PALT is a laser altimeter/rangefinder that measures line-of-sight distance to the asteroid surface using time-of-flight of 1550 nm laser pulses.

- **Performance:** nominal 10 m to 20,000 m range, 0.1 m resolution.
- **Multi-target:** up to five targets per line of sight.
- Measurements are available via the data pool, refreshed every two seconds, and can be read using `Hera_Read_Parameter_float64()`.

3. OUTPUTS (TELEMETRY AND REPORTING)

The Experimental Software (ESW) can generate outputs only through standard CCSDS / ECSS PUS telemetry services, using the platform API. The on-board software on Core 0 receives these requests, builds the telemetry packets, and stores them in spacecraft mass memory for later downlink, within the limits below.

Proposers should treat output generation as a core part of the experiment design: outputs are how the experiment is monitored and used operationally.

3.1. Housekeeping Telemetry — PUS Service 3

Housekeeping (HK) is the standard monitoring service in ECSS PUS. For ESW, HK is the main way for operators to confirm correct execution and track key status information (e.g. mode, counters, progress, health).

- **Minimum interval:** at least 5 minutes between HK reports.
- **Maximum size:** 256 bytes per HK packet.

If your experiment is selected and uses HK, your ICD documentation shall specify:

- the HK report structure
- the list of reported parameters, including offset, type, size, and calibration
- the total HK data volume per execution window (in kBytes)

Recommended practice: keep HK limited to a small set of stable indicators (mode, cycle counter, last-result code, summary metrics). Do not try to stream high-rate data via HK, as it will not fit the size and interval limits.

3.2. Event Reporting — PUS Service 5

Event Reporting is the ECSS PUS notification service. It is used to inform Ground when important events occur, such as completion of a major processing step, a threshold crossing, a detection, or a mode change.

- **Minimum interval:** at least 20 seconds between event reports.
- **Maximum size:** 50 bytes per event packet.

Only two severity levels are allowed for ESW:

- Informational
- Warning

If your experiment is selected and uses events, your ICD documentation shall specify:

- the event structure (event identifiers and layout)
- the event parameters, including offset, type, size, and calibration (as applicable)

- the total event data volume per execution window (in kBytes)

Recommended practice: Use events for meaningful, discrete conditions (start/stop, completion, detection, recoverable anomalies). Avoid repetitive “heartbeat” events—those belong in HK and must still respect the HK minimum interval.

3.3. Science Data Telemetry

Science Data telemetry is used to downlink experiment results, such as processed image products, extracted features, summaries, or compact descriptors.

- **Maximum size:** 2048 bytes per packet (including headers).
- **Maximum volume per 3 hour slot**: 12 MB.

If your experiment is selected and uses Science Data telemetry, your ICD documentation shall specify:

- the data format (structure/content of the payload, including any compression or encoding)
- the maximum science data volume per execution window

4. DEVELOPMENT PROCESS AND SOFTWARE PRODUCT ASSURANCE

A key part of this call is compliance with European space standards. The Core 1 sandbox makes it safer to fly experimental software, but it does not remove the need for a solid engineering process. The main standards are ECSS E ST 40C (software engineering) and ECSS Q ST 80C (software product assurance).

Even though the ESW is Category D, selected experiments must still follow a clear development and verification approach to ensure the selected software performs as expected and can be reviewed, integrated, and operated reliably.